

FloodGap - A sensor-siting model inspired by Savannah's CEAR Hub, applied to Houston

VIP: Coastal Georgia Climate Solutions, Georgia Tech OMSCS, Spring 2026

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Products: [GitHub repository](#)

1. Summary

This research project was completed for Georgia Tech's OMSCS program in Spring 2026. It began with a question: if Houston had funding for 20 new water-level sensors, where should they go? My initial approach was to build a weighted scoring model, calibrate it against CEAR Hub's 48 existing sensors in coastal Georgia, and then transfer the model to Houston to identify the top 20 locations.

Working with the Georgia data reshaped that approach. The existing sensors had been sited by installation feasibility rather than by ideal placement, which meant they could not serve as a ground truth for calibrating weights. Rather than fit the model to a target that did not exist, I redesigned the method around that constraint: I treated the weights as explicit policy options, scored 1,211 Houston-Galveston census tracts under five distinct weighting schemes, and identified the locations that rank highly under every scheme.

This robustness-based approach yields four priority clusters: Texas City / La Marque, Gulfton / Sharpstown, East End / Ship Channel, and Kashmere / Fifth Ward, each matched to an appropriate sensor technology (water-level gauges versus street-light ultrasonic units) and paired with a field-survey method for translating tract-level results into specific poles and bridges.

2. Starting point (Weeks 1-2)

The idea came from CEAR Hub's sea-level sensor network around Savannah. Watching real-time water-level data feed flood awareness made me ask whether the same approach could transfer to other cities. I live in Houston, which floods in several distinct ways, so the question localized naturally.

Two framings were on the table early: (A) stay entirely in coastal Georgia, where sensor density gives the strongest ground truth, or (B) validate the method in Georgia and then apply it to Houston-Galveston, where flood days are rising among the fastest in the country but the local sensor network is sparse. I chose B. With Professor Clark's guidance the question became concrete:

If Houston had funding for 20 new sensors, where are the 20 most ideal places to put them?

Context numbers I collected at this stage: high-tide flooding days across the U.S. Southeast are up roughly 500% since 2000. The upper Texas coast (including Galveston) is up over 400%.

3. First model and the first problem (Weeks 2-3)

I started with two layers: [NOAA high-tide flood days](#) and [CDC/ATSDR Social Vulnerability Index \(SVI\)](#), joined to census-tract locations from the TIGER gazetteer by FIPS code.

Problem: NOAA flood days alone. NOAA tide stations are so sparse that all 174 coastal Georgia tracts inherited the same flood-day value (9 days) from their nearest station. A factor with zero variation cannot rank anything. I spent time confirming this was not a bug in my join before accepting that it is simply what sparse station data looks like at community scale.

Solution: add USGS elevation (via the Open-Elevation API). Low-lying ground floods first, and elevation varies tract by tract, so it restores exactly the local signal NOAA cannot provide. The flood factor became NOAA days + elevation together.

This produced the first durable lesson of the semester, which shaped every later data decision: the data has to vary at the scale you are ranking. It is also the argument for the whole project: sparse regional monitoring cannot tell one neighborhood from another, which is why a denser sensor network, and the question of where to put it, matters at all.

At this stage the score was: $\text{score} = \text{low-lying} \times 0.4 + \text{vulnerability} \times 0.4 + \text{flood days} \times 0.2$, with the weights explicitly marked as placeholders to be replaced by something data-grounded later.

4. The reference paper and the validate-then-apply plan (Week 3)

Kait shared [Tien, Lozano and Chavan \(2023\), Communications Earth & Environment](#), a study from a previous PhD student in the group on siting sensors in the exact same Chatham County network. Their approach: five parameters (compound flood exposure, social vulnerability, infrastructure density, coverage of the existing network, uncertainty reduction) computed at ~28,890 candidate locations, narrowed by Pareto non-dominated sorting to 381 points (about 1.3%), with local experts and county officials making the final calls. Notably, they deliberately avoid scalar weights. The trade-offs are left to decision-makers. Also notably, the infrastructure priorities in the paper were set by Chatham Emergency Management (CEMA), not by the researchers, which foreshadowed something I would learn later about how much of siting is institutional rather than algorithmic.

My plan borrowed their factor set but used weighted scoring instead of Pareto, because I wanted a ranked top-20 rather than 381 candidates needing expert triage. The plan was:

1. Build the weighted-score model from their factors.
2. **Validate** it against CEAR Hub's existing sensor locations: if the model scores high where experts put sensors, it captures expert judgment.
3. If it works in Savannah, **apply** it to Houston.

5. Validation, and the second problem (Weeks 5-6)

I obtained the 48 CEAR Hub sensor locations from Professor Clark and scored them with the model.

Problem: the validation itself. Under my placeholder weights, the 48 sensor sites averaged only the 42nd percentile of the model's score distribution, slightly below random. Early cuts were even starker: only 2 of 48 sensors fell in my top 10% zone, 4 in the top 20%.

I tried to rescue the validation by reverse-inferring weights from the sensors (feature comparison: each parameter's mean at sensor sites vs. the regional mean). That exercise was informative in an unexpected way: every weight combination that fit the sensors well pushed the social-vulnerability weight toward zero, and flood + elevation together jumped agreement dramatically (top 10% hit rate from 4% to ~50%; top50% from 40% to ~73%). In other words, the experts sited almost purely on tidal flood physics and feasibility, not on vulnerability.

Then Professor Clark explained how the 48 sensors were placed, which resolved the puzzle completely:

- Installed 6-8 years ago, chosen for **feasibility**: easy access, near a waterway, something to mount on (often literally a fence post).
- Theft and mounting stability were real constraints.
- The deployment was never fully completed, due to county infrastructure limits.
- Siting was driven almost entirely by tidal flood risk.

So the 48 sensors are "where it was possible to install," not "where it is ideal to install." Using them as ground truth to fit weights is not valid, not because the model is wrong, but because no ground truth of ideal placement exists.

Solution: The model deliberately scores what should matter. The gap between the two is precisely the model's contribution, it surfaces high-risk, vulnerable areas the current network systematically misses. This is also the second meaning of the project's name: FloodGap finds the gap between where sensors are and where they should be.

Professor Clark also distilled the real siting process into three questions I kept for the rest of the project: Where is the flood? Who is impacted? Can a sensor technically work there? And the paper's own text confirms the institutional side: the Chatham work was a collaboration with county officials, with CEMA choosing which infrastructure counted. Siting is expert-and-official-led with algorithmic support, not algorithm-led.

6. Transfer to Houston, and the third problem (Weeks 6-7)

Problem: applying the Savannah-style model to Houston unchanged. Savannah's flooding is more than 90% tidal, so a tidal flood factor is a reasonable stand-in for flood risk there. Houston floods three ways: tidal surge near Galveston Bay, stormwater ponding inland where rainfall exceeds drainage, and riverine flooding along the bayous. Running the tidal-only model on the 1,211 Houston-Galveston tracts put essentially every priority site on the coast: Harris County, with 1,110 tracts, landed only 3 sites in the top 50, while Galveston County's 101 tracts took 47. Galveston tracts entered the top 50 at roughly 30x the per-tract rate of Harris. Harvey's catastrophe was inland stormwater; the model was blind to exactly the flooding Houston is known for.

Solution: a compound flood factor with three normalized sub-components:

- **Tidal:** [NOAA flood days](#) + [USGS elevation](#) (as in Georgia).
- **Stormwater:** [impervious surface fraction from NOAA C-CAP high-resolution land cover](#), sampled from the 30 m raster in a ~1 km window at each tract center, weighted by NOAA's developed-class impervious coefficients. Pavement is what turns rain into street flooding.
- **Riverine:** distance to the nearest of 11 major bayous/creeks (Buffalo, Brays, White Oak, Greens, Sims, Halls, San Jacinto, Cypress, Clear, Dickinson, Chocolate), inverted so closer = higher risk.

With equal sub-weights, Harris went from 3 to 45 of the top 50. The inland risk finally showed up in the map.

Smaller things that did not work along the way (Weeks 4-7):

- **OpenStreetMap/Overpass as a live data source.** I originally planned OSM for building density (stormwater proxy), waterways (riverine), and critical infrastructure. The Overpass API timed out constantly at the query sizes I needed. I replaced building density with the C-CAP raster (better data anyway, actual impervious cover, not a proxy), replaced OSM waterways with the hardcoded 11-bayou set, and effectively dropped the infrastructure factor to zero weight. The lesson: pipelines that depend on a flaky external API are not pipelines; the final model runs fully offline from downloaded data.
- **A county-name regex bug.** My Texas SVI filter for "Harris" also matched "Harrison County" (which is near Louisiana, nowhere near Houston), silently inflating the study area to 1,227 tracts. Caught it when a mapped point plotted far outside the region; fixed the filter; final count 1,211 (Harris 1,110 + Galveston 101).
- **Block-group SVI.** Kait noted that tract-level CDC SVI under-counts marginalized and immigrant populations and pointed me to finer indices. Computing block-group SVI requires the R package `findSVI` plus a Census API key. I ran out of time before doing it properly. It is listed as future work rather than silently ignored. The current model uses tract-level SVI with this limitation stated.

7. Solution: weights as policy options (Week 7)

Problem: By this point the weight question had become unavoidable. My chain of worry was: the Georgia validation failed, so reverse-inferred weights are unreliable, so my weights are "inaccurate", and now I am adding two new sub-components whose weights have even less basis.

The resolution was realizing the premise was wrong. The reference paper avoids weights on purpose (Pareto). The experts' own placements encode feasibility, not a weighting of risk factors. The Harris County gauge-placement literature uses network observability, a different formalism with no weights either. There is no true weight vector to be inaccurate relative to.

Solution: So I stopped treating weights as parameters to estimate and started treating them as policy options to expose. If a decision-maker prioritizes tidal surge, the model gives one top 20; if stormwater, another; if vulnerable communities, another. The model's job is to show the trade-offs honestly and to identify the sites that

stay on top under every reasonable priority: the robust sites. Methodologically this is a sensitivity analysis; practically it matches how siting decisions are actually made.

8. Five weighting schemes (Week 8)

I defined five schemes. Internal weights split the compound flood (tidal/storm/river); final weights split flood vs. vulnerability:

Scheme	tidal/storm/river	flood/vuln	Harris/Galv in top 50	Leading regions
A Tidal-only	100/0/0	60/40	3 / 47	Dickinson, Texas City, Galveston
B Equal	33/33/33	60/40	45 / 5	Gulfton, East End, Texas City
C Harvey-informed	25/50/25	60/40	48 / 2	Gulfton, East End, Texas City
D Flood-dominant	33/33/33	80/20	42 / 8	Gulfton, East End, Texas City
E Equity-forward	33/33/33	50/50	46 / 4	Gulfton, Texas City, East End

Overlaying all five: only **2 of 1,211 tracts** are chosen by every scheme (both in Texas City / La Marque); 28 more appear in 4 of 5; 14 at 3/5; 17 at 2/5; 52 at 1/5. In total 113 tracts reach a top-50 under at least one scheme. The robust top 20 (the 2 unanimous sites plus the 18 best at 4/5) is the project's headline recommendation: 18 Harris / 2 Galveston, with 8 of the 20 in Gulfton.

What did not work: the coverage factor. Following the paper, I had included a coverage term (normalized distance from the risk-weighted centroid, 10-12% weight) meant to keep the 20 sensors from clustering. When I finally audited it, it failed on both ends: it did not prevent clustering (the top 20 still contained 25 pairs of sites within one mile of each other, minimum spacing 0.18 mi), and it correlated negatively with the risk signals it shared a score with ($r = -0.61$ vs. riverine, -0.45 vs. stormwater, -0.21 vs. SVI), effectively rewarding low-risk periphery. Removing it made the five-scheme comparison cleaner (the tidal-only to Harvey contrast sharpened from 3-45 to 3-48) and cost nothing. The honest fix for spatial spread is not a factor inside the score at all but a post-ranking greedy minimum-separation selection; that is implemented conceptually and listed in next steps. Removing coverage also changed one number I had been quoting: Gulfton's rank jump under compound weighting went from " ~ 430 to $\sim 54-88$ " to " ~ 400 to $\sim 111-143$ ". Less dramatic, more truthful.

9. Zooming into the clusters (Weeks 9-10)

The robust tracts organize into four clusters, each with a distinct character and, importantly, a distinct sensor technology implied by its dominant flood component:

- **Texas City / La Marque.** The robust core; the only region in all five schemes. Tidal component 0.92 (highest surge exposure), SVI 0.69, behind the levee/dike system. Sensor: **water-level** on bay-front bridges, the causeway and dike structures.
- **Gulfton / Sharpstown.** The compound champion; top region in 4 of 5 schemes (16-23 of the top 50, rank #1 under Harvey-informed weights). A 1960s-70s garden-apartment district, now among Houston's densest neighborhoods; 65% impervious (highest in the study area), SVI 0.74, majority-renter immigrant communities in the Brays Bayou watershed; flooded in Allison '01, Memorial Day '15, Tax Day '16, Harvey '17. Invisible under tidal-only weights (median rank ~ 400), exactly the inland risk the coastal logic missed. Sensor: **street-light ultrasonic** (pole-mounted, measures street ponding, no waterway needed; NYC FloodNet model, a few hundred dollars each).

- **East End / Ship Channel.** Closest to water of any cluster (0.6 mi to the bayou), SVI 0.82, industrial-adjacent (Manchester, Magnolia Park), a long-standing environmental-justice concern. Sensor: **mixed**, water-level on the Buffalo Bayou / Ship Channel structures plus street-light units inland.
- **Kashmere / Fifth Ward.** The highest social vulnerability in the study area (SVI 0.91), on Hunting Bayou, 54% impervious; chronic street flooding meets the least capacity to recover. Sensor: **street-light**, plus a water-level unit at the Hunting Bayou Road crossings.

Each recommended site in the output carries a sensor-type tag derived from its dominant flood component. This closes the loop back to Professor Clark's third siting question (can a sensor technically work there): the two technologies have different physical requirements, and the model now speaks to both.

For each cluster I also drafted a two-phase field-survey method (a desktop pre-screen using Street View, 311 flooding complaints, HCFCD flood history, pole/bridge ownership and cellular coverage, followed by an on-site checklist covering street low points, stain lines, storm inlets, mounting structures, base-flood-elevation clearance, permits and access). Site reconnaissance is my day job as a landscape architect, so this is the part of the pipeline where I can contribute something the model cannot.

10. Final results

All files are in the [GitHub repository](#).

Code

- [floodgap_compound.py](#)
- [render_maps.py](#)

Interactive

- [floodgap_weight_explorer_satellite.html](#)
- [floodgap_five_schemes_satellite.html](#)

Presentations

- [Final presentation deck](#)

Data sources (external)

- NOAA CO-OPS high-tide flooding / station data: <https://tidesandcurrents.noaa.gov/>
- USGS elevation via Open-Elevation API: <https://open-elevation.com/>
- CDC/ATSDR SVI: <https://www.atsdr.cdc.gov/placeandhealth/svi/>
- NOAA C-CAP high-resolution land cover: <https://coast.noaa.gov/digitalcoast/data/ccaphighres.html>
- Census TIGER gazetteer: <https://www.census.gov/geographies/reference-files.html>
- Reference paper: Tien, Lozano & Chavan (2023), Comm. Earth & Environ. 4:96: <https://www.nature.com/articles/s43247-023-00761-1>
- NYC FloodNet (street-light sensor model): <https://www.floodnet.nyc/>

11. Limitations, stated plainly

- The compound-flood internal weights cannot be learned from Savannah (no stormwater/riverine variation there to learn from). They are policy choices, which is why the five-scheme sensitivity structure exists. This is the model's honest core limitation, and it is presented as such rather than hidden.
- Tract-level SVI under-represents marginalized populations; block-group SVI is the known upgrade.
- The riverine factor uses straight-line distance to 11 simplified bayou centerlines, not hydraulic modeling or FEMA floodway geometry.
- The stormwater factor is impervious fraction only; it does not yet incorporate drainage capacity, storm-sewer condition, or observed ponding records (311/HCFCD data are planned for the block-level refinement).

- Scores rank tracts, not exact installation points; the field-survey phase exists precisely to bridge that gap.
- The scale bars on zoom maps are EPSG:5070 metric distances; adequate for presentation, would be re-derived for any engineering use.

12. Next steps

- **Fine-grain inside each Houston cluster.** Re-run scoring at block-group level within Gulfton, Texas City, East End and Kashmere; rank streets and intersections using 311 flooding complaints and HCFCD flood history.
- **Desktop pre-screen of candidate sites.** Street View + drainage records to shortlist 3-5 exact poles or bridges per cluster; check ownership (CenterPoint, TxDOT, HCFCD) and cellular signal.
- **Field reconnaissance.** Structured site visits with the checklist; photograph low points, stain lines, inlets and mounting structures.
- **Test against known gaps.** Compare model picks with the areas the team wanted to monitor but could not install in, and share results with Houston emergency management for real-world feedback.
- **Minimum-separation selection.** Replace the removed coverage factor with post-ranking greedy spacing.
- **Block-group SVI + FEMA flood zones.** The two known data upgrades.
- **Transferability test.** Point the same pipeline at another region (e.g., Florida) to see whether the cluster-finding method generalizes; longer-term venues: SCDRP annual meeting (March) and the Georgia Resilience Conference.